

✓ Exploratory Data Analysis (EDA) on the Titanic Dataset

Objective: Analyze the Titanic dataset using Python to identify trends, patterns, relationships, and insights through statistical analysis and data visualization.

Tools Used:

- Python
- Pandas
- Matplotlib
- Seaborn

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Display plots inside the notebook
%matplotlib inline

# Set plot style
sns.set_style("whitegrid")
```

```
# Load the Titanic dataset
df = pd.read_csv("train.csv")

# Display the first 5 rows
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599
				Heikkinen, STON/O2					

Next steps:

Generate code with df

New interactive sheet

```
# Dataset dimensions
print("Dataset Shape:", df.shape)

# Column names
print("\nColumns:")
print(df.columns)

# Dataset information
print("\nDataset Information:")
df.info()
```

Dataset Shape: (891, 12)

Columns:

```
Index(['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp',
       'Parch', 'Ticket', 'Fare', 'Cabin', 'Embarked'],
      dtype='object')
```

Dataset Information:

```
<class 'pandas.core.frame.DataFrame'>
```

RangeIndex: 891 entries, 0 to 890

Data columns (total 12 columns):

#	Column	Non-Null Count	Dtype
0	PassengerId	891 non-null	int64
1	Survived	891 non-null	int64
2	Pclass	891 non-null	int64
3	Name	891 non-null	object
4	Sex	891 non-null	object
5	Age	714 non-null	float64
6	SibSp	891 non-null	int64
7	Parch	891 non-null	int64
8	Ticket	891 non-null	object
9	Fare	891 non-null	float64
10	Cabin	204 non-null	object
11	Embarked	889 non-null	object

dtypes: float64(2), int64(5), object(5)

memory usage: 83.7+ KB

Observation

- The Titanic dataset contains passenger details such as age, gender, ticket class, fare, and survival status.
- Some columns contain missing values that will be explored later.
- The dataset includes both numerical and categorical features for analysis.

```
# Statistical summary of numerical columns
df.describe()
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch	
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	89
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	3
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	4
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	1
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	3
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	51

Observation

- The statistical summary provides an overview of numerical features such as Age and Fare.
- There is noticeable variation in passenger ages and ticket fares.

# Check missing values df.isnull().sum()	
	0
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2
dtype: int64	

Observation

- Missing values are present in Age, Cabin, and Embarked
- Cabin has the highest number of missing values.

```
print("Passenger Class")
print(df["Pclass"].value_counts())

print("\nGender")
print(df["Sex"].value_counts())

print("\nSurvival")
print(df["Survived"].value_counts())
```

```
Passenger Class
Pclass
3      491
1      216
2      184
Name: count, dtype: int64
```

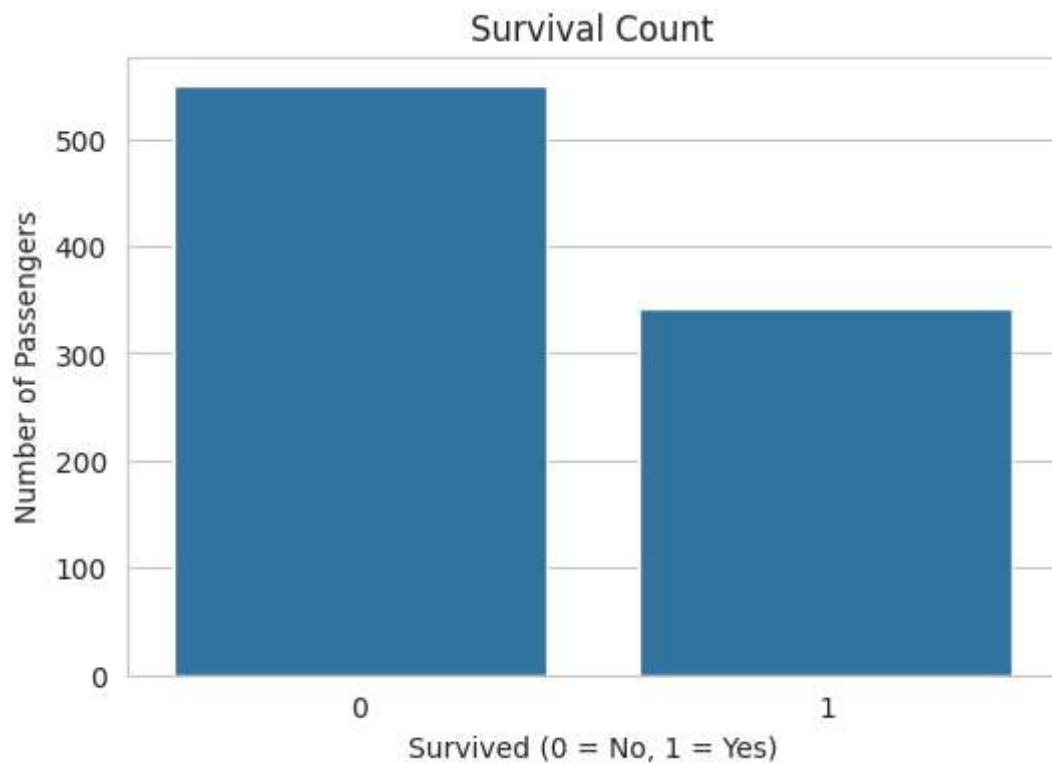
```
Gender
Sex
male      577
female    314
Name: count, dtype: int64
```

```
Survival
Survived
0      549
1      342
Name: count, dtype: int64
```

Observation

- Most passengers traveled in Third Class.
- Male passengers outnumber female passengers.
- More passengers did not survive than survived.

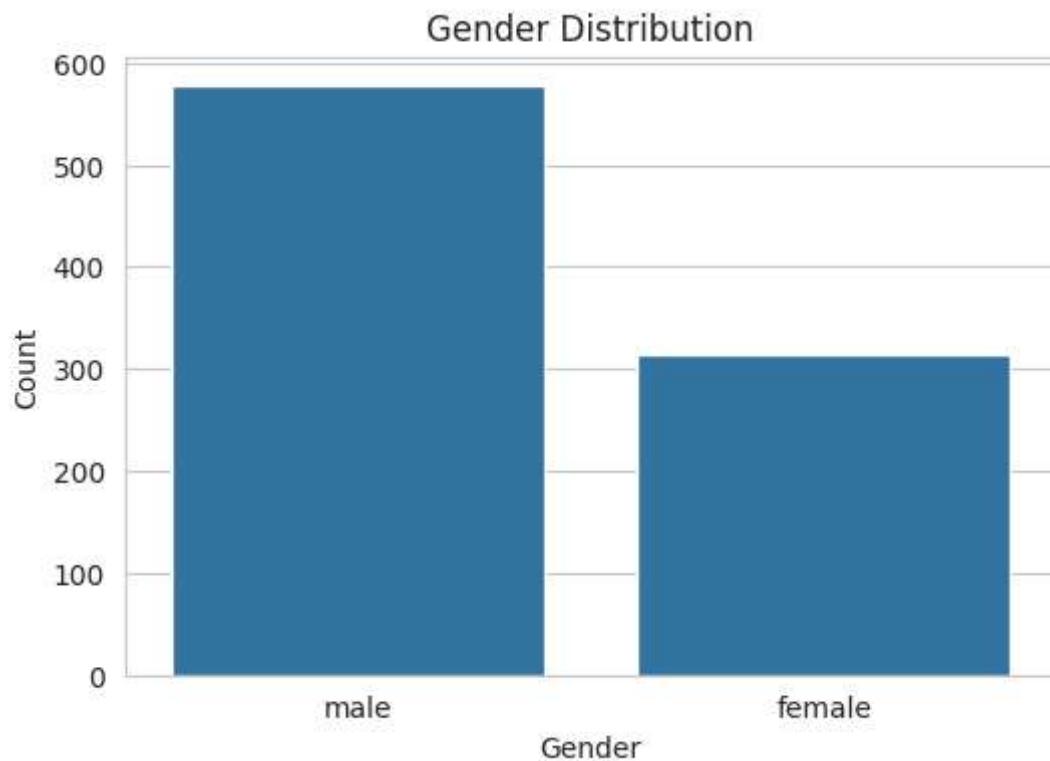
```
plt.figure(figsize=(6,4))
sns.countplot(x='Survived', data=df)
plt.title("Survival Count")
plt.xlabel("Survived (0 = No, 1 = Yes)")
plt.ylabel("Number of Passengers")
plt.show()
```



Observation

- More passengers did not survive than survived, indicating the disaster had a high mortality rate.

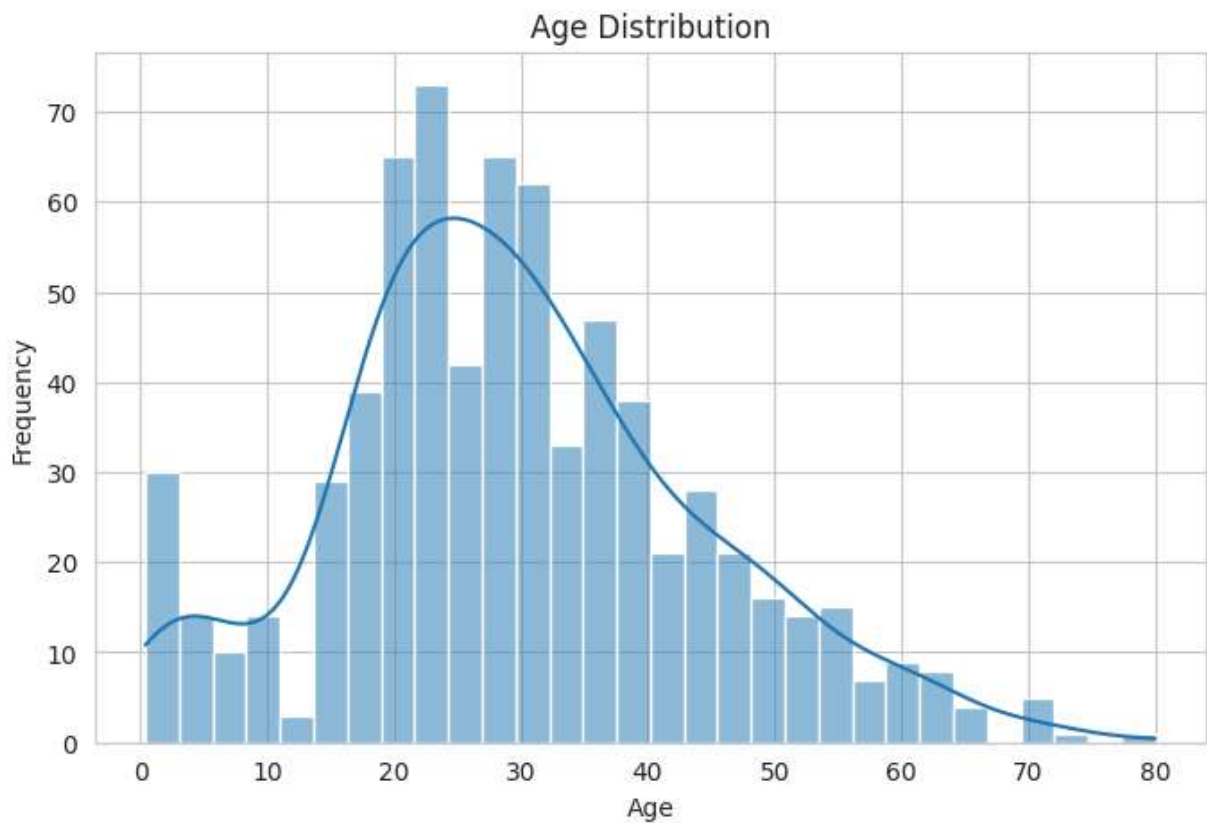
```
plt.figure(figsize=(6,4))
sns.countplot(x='Sex', data=df)
plt.title("Gender Distribution")
plt.xlabel("Gender")
plt.ylabel("Count")
plt.show()
```



Observation

- The dataset contains more male passengers than female passengers.

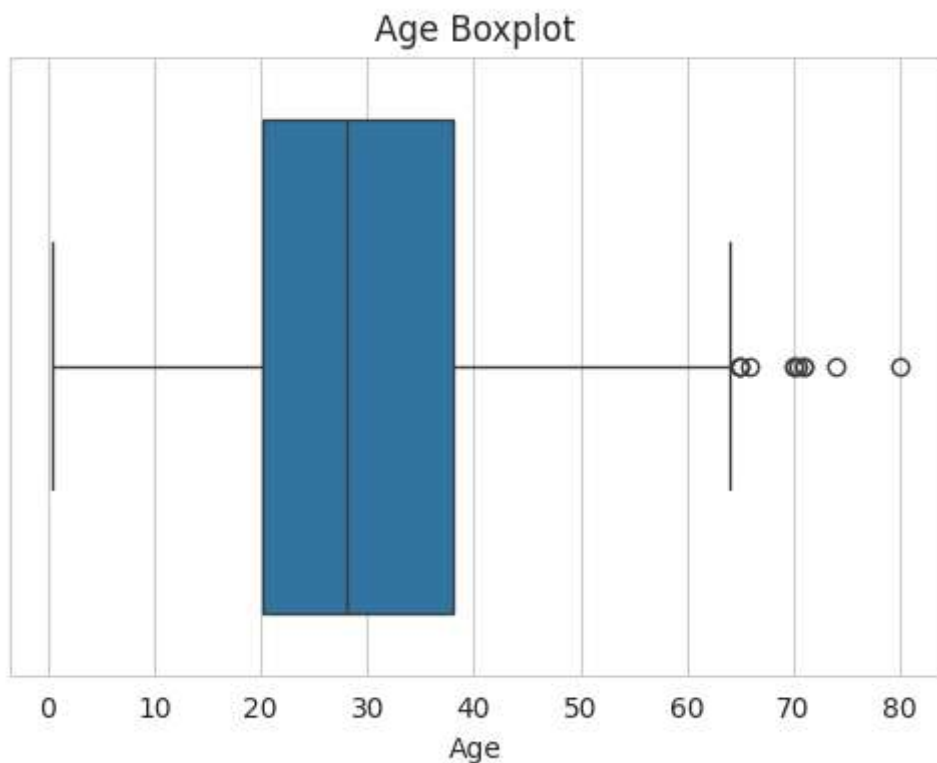
```
plt.figure(figsize=(8,5))
sns.histplot(df['Age'], bins=30, kde=True)
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Frequency")
plt.show()
```



Observation

- Most passengers were between 20 and 40 years of age, with fewer elderly passengers.

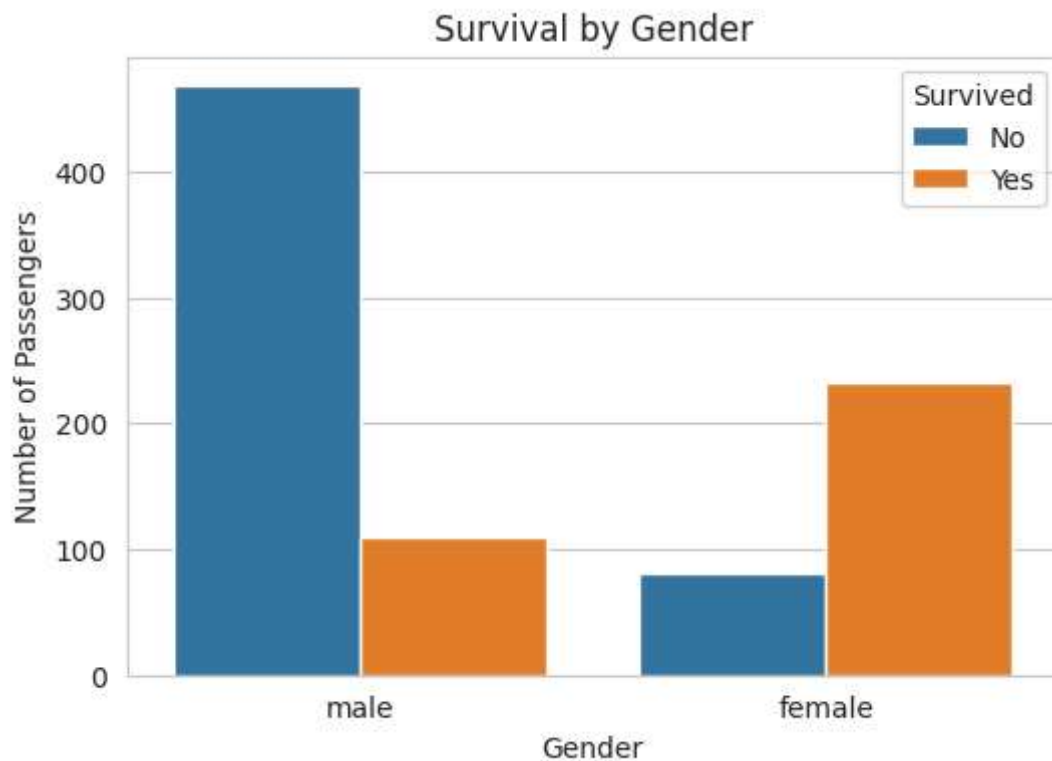
```
plt.figure(figsize=(6,4))
sns.boxplot(x=df['Age'])
plt.title("Age Boxplot")
plt.show()
```



Observation

- The age distribution contains several outliers, representing unusually old or young passengers.

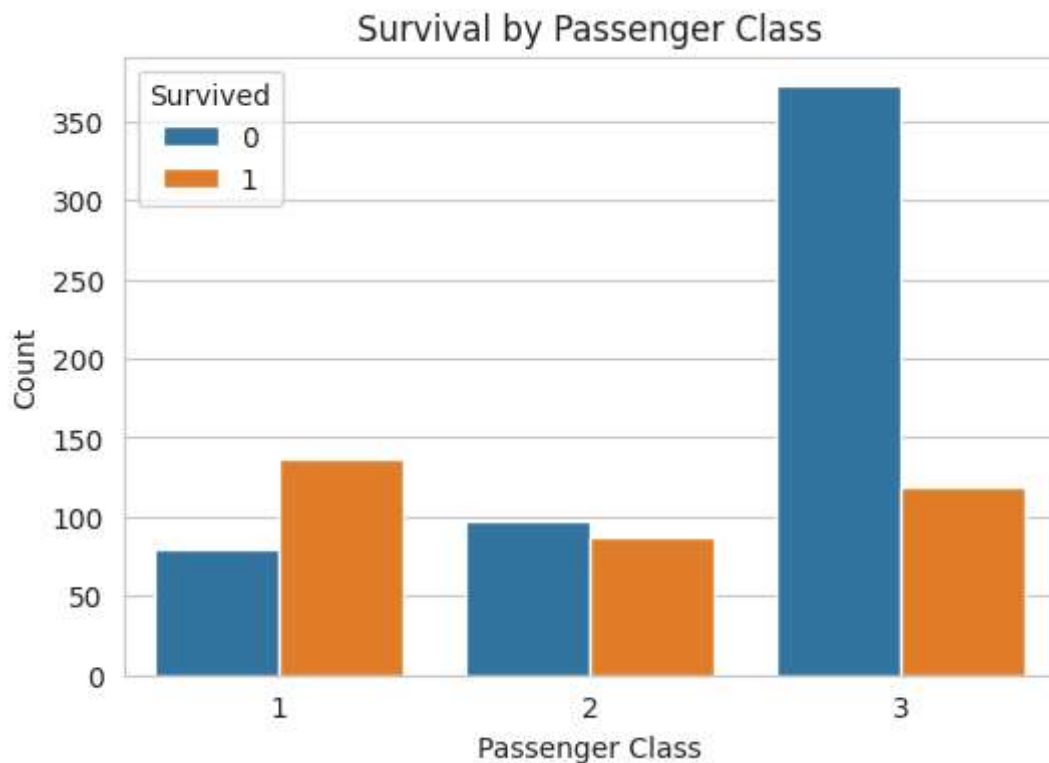
```
plt.figure(figsize=(6,4))
sns.countplot(x='Sex', hue='Survived', data=df)
plt.title("Survival by Gender")
plt.xlabel("Gender")
plt.ylabel("Number of Passengers")
plt.legend(title="Survived", labels=["No", "Yes"])
plt.show()
```

Observation

- Female passengers had a significantly higher survival rate than male passengers.

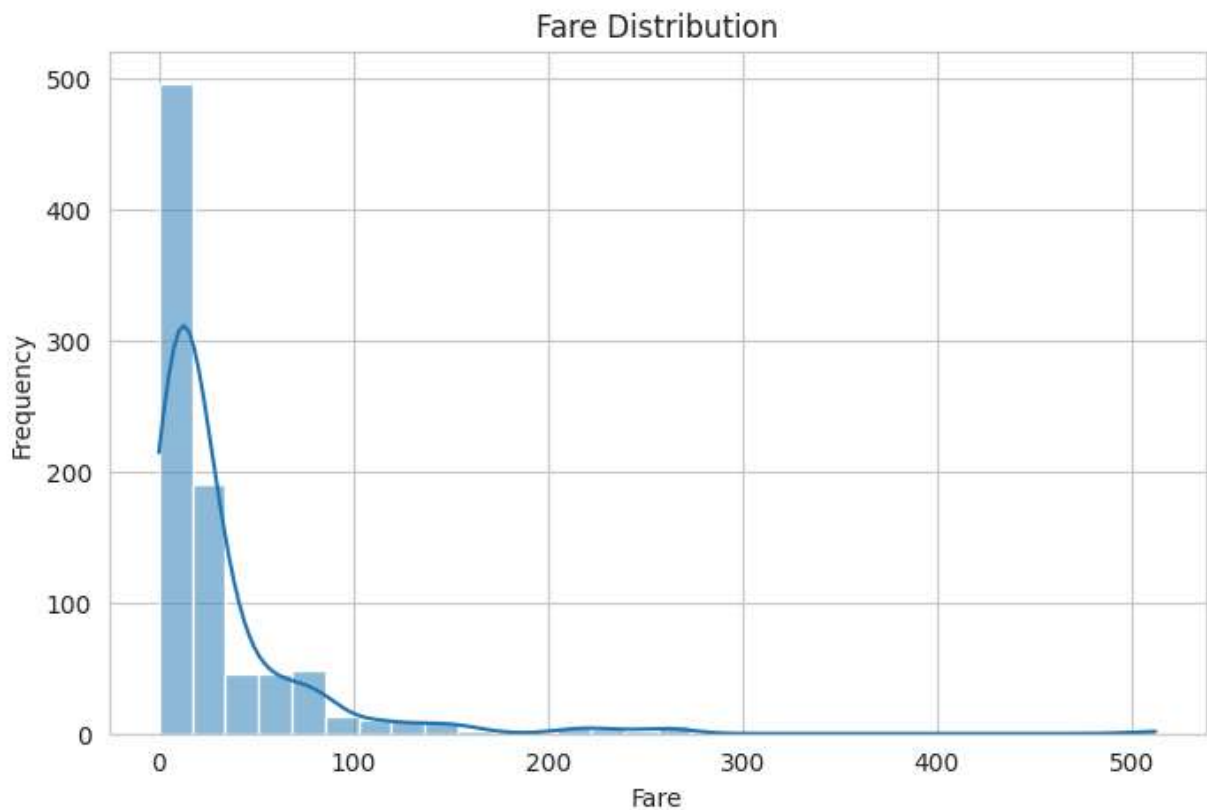
```
plt.figure(figsize=(6,4))
sns.countplot(x='Pclass', hue='Survived', data=df)
plt.title("Survival by Passenger Class")
plt.xlabel("Passenger Class")
plt.ylabel("Count")
plt.show()
```



Observation

- First-class passengers had the highest survival rate, while third-class passengers had the lowest.

```
plt.figure(figsize=(8,5))
sns.histplot(df['Fare'], bins=30, kde=True)
plt.title("Fare Distribution")
plt.xlabel("Fare")
plt.ylabel("Frequency")
plt.show()
```



Observation

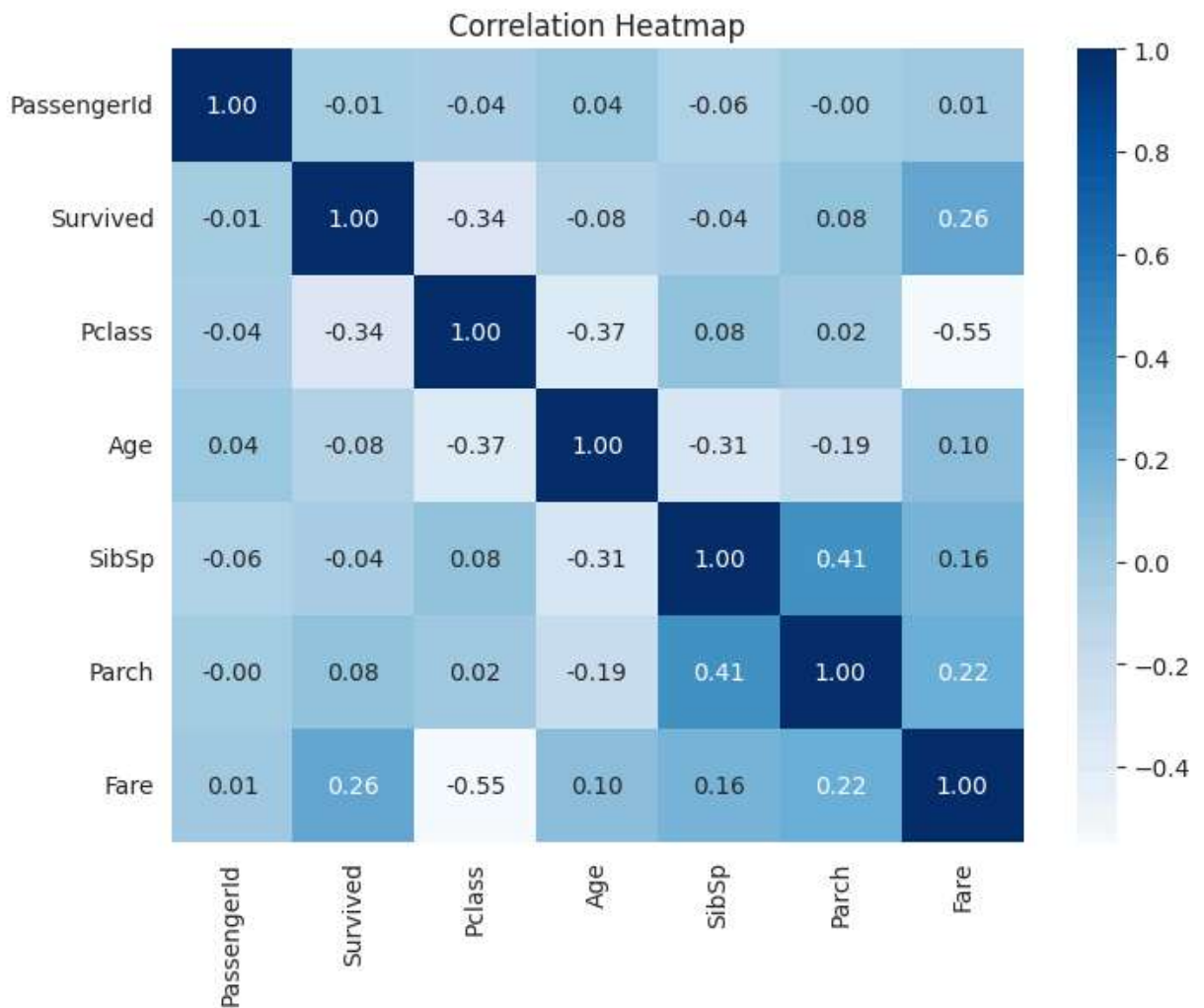
- Most passengers paid lower fares, while only a few passengers paid very high ticket prices.

```
plt.figure(figsize=(8,6))

numeric_df = df.select_dtypes(include=['number'])

sns.heatmap(numeric_df.corr(), annot=True, cmap='Blues', fmt=".2f")

plt.title("Correlation Heatmap")
plt.show()
```



Observation

- Passenger class and fare show relationships with survival, while some variables have only weak correlations.

```
sns.pairplot(df[['Survived', 'Age', 'Fare', 'Pclass']].dropna(), hue='Survived')
plt.show()
```

